

Cian O'Toole

📍 Dublin, Ireland ✉ cotoole3@tcd.ie 🌐 My Website [🔗](#)

Education

Trinity College Dublin

Sept 2023 – Sept 2027

PhD in Astrophysics: Weather on Worlds Beyond our Own.

I research the drivers of variability in early L-dwarfs, disentangling auroral, cloud and magnetic spot driven variability. I have a keen interest in exo-aurorae, in the ultraviolet, near-infrared and the radio.

Supervisor: Prof. Johanna Vos

Trinity College Dublin

Sept 2019 – May 2023

BA in Physics and Astrophysics

- Grade: 2.1

First and Senior Author Publications

- *Disentangling Auroral, Cloud and Magnetic Spot Driven Variability in Three Early L-dwarfs with HST/WFC3*
O'Toole et al. [Astronomy & Astrophysics](#) [🔗](#).
- *HST+VLA Observations of the T6.5 dwarf 2MASS 1047+21*
O'Toole et al. (in prep)
- *Atmospheric Variability and Spin Axis Alignment of the Planetary-Mass Companion BD+60B*
Vos, **O'Toole**, Lam et al. (in prep)

Nth Author Publications

- *The JWST Weather Report from the Isolated Exoplanet Analog SIMP 0136+0933: Pressure-Dependent Variability Driven by Multiple Mechanisms*, The Astrophysical Journal Letters
Allison McCarthy, Johanna Vos, Philip Muirhead et al.; + 17 other co-authors, incl. **O'Toole, C.**
- *The JWST Weather Report: auroral heating and patchy clouds on SIMP J0136 measured with time-resolved atmospheric retrievals.*, Astronomy and Astrophysics
Evert Nasedkin et al, incl. **O'Toole C.**
- A Jupiter-like radio aurora in the L3.5 dwarf LSPM J0036+1821
Martín-Carrero et al, incl. **O'Toole C.** (in prep)

Telescope Time

- **HST Cycle 33:** *UV Exo-Aurorae: Multi-Wavelength Observations of a Radio Emitting T-dwarf to confirm Auroral Activity in the Ultraviolet*, PID 18113, **6 orbits**, with **9 hours** of simultaneous **VLA** observations.
PI: O'Toole C
- **UKIRT: DDT:** *Novel simultaneous monitoring of a radio-emitting brown dwarf.* **Half night.**
Co-PI: O'Toole C
- **CFHT: 2026B:** *Bridging the Interior and Atmosphere: NIR Variability Search of Radio-Periodic Brown Dwarfs.* **8.9 hours.**
PI: Xueqing Chen
Co-I: O'Toole C
- **JWST Cycle 4:** *Storms are Brewing in the ONC: Variability Monitoring of 50 Substellar Worlds*, PID 8155.
PI Limbach M, Co-PI Vos J
Co-I: **O'Toole C**

Talks

- Invited talk at the UK Exoplanet Meeting 2026, University of Bristol, April 2026
- Contributed talk at Other World's Laboratory, UCSC, California, USA, July 2025
- Contributed Talk at European Astronomical Society Conference, Cork, Ireland, June 2025
- Invited Talk at the DUBlin STar formation, deBris dIsks and plaNets (DUSTBIN) meeting, October 2024, March 2026.
- Contributed Talk at Irish National Astronomical Meeting, Galway, Ireland, August 2024.
- Contributed Talk at Dublin-Belfast Exoplanet meeting, August 2024.

Posters

- Cool Stars 23, Tokyo, Japan.
- Institute of Physics Ireland Spring Conference, Galway, Ireland, February 2025
- Carl Sagan Summer School, July 2024.
- Cool Stars 22, San Diego, California, USA, [June 2024](#) [🔗](#).

Grants and Awards

- NASA JPL travel grant, Cool Stars 22, \$900, June 2024.
- Runner up in TCD School of Physics Tercentenary *3 Minute Thesis* competition, €100, April 2024.

Outreach

- Talk at Pint of Science, Dublin, May 2026
- Irish Astronomy Week Outreach talk at Kevin Street Library, March 2026.
- Space Week talk at Explorium, Dublin, Ireland, October 2025
- Volunteer at EAS 2025, Cork, Ireland.
- Talk at Pint of Science, Dublin, May 2025
- Space Week talk at Explorium, Dublin, Ireland, [October 2024](#) [🔗](#)
- SFI Curious Minds for National Science Week, [October 2024a](#) [🔗](#), [October 2024b](#). [🔗](#)
- TCD Astrophysics Seminar Organiser, September 2024 - present.
- TCD School of Physics Annual Charity Fundraiser Organizer, 2022 - present (€4,200 raised).

Teaching

- Teaching Assistant in the 3rd year undergraduate experimental labs, September 2023 - Present